

YASHRAJ AGAWANE

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Professional Summary

Software Developer and Machine Learning Enthusiast with strong foundations in Java, Python, C++, and Data Structures. Experienced in building AI-powered applications, full-stack web development, and real-time systems using modern frameworks, REST APIs, and machine learning techniques. Seeking opportunities to apply problem-solving and software engineering skills in impactful projects.

Skills

Programming: Java, Python, C++, C, SQL

Web Technologies: Next.js, React, HTML, CSS, JSP, .NET

Machine Learning: Scikit-learn, XGBoost, Data Preprocessing, Feature Engineering, Model Evaluation

Tools: Git, GitHub, Streamlit, Render, Vercel, VS Code

Concepts: Data Structures and Algorithms, OOP, DBMS, REST APIs, Computer Vision, Operating Systems

Projects

Driver Drowsiness Detection System | *Python, Flask, OpenCV, Dlib, Computer Vision*  **Nov 2025 – Jan 2026**

- **Developed** a real-time computer vision monitoring system leveraging **OpenCV** and **Dlib** to compute Eye Aspect Ratio (EAR) through facial landmark detection for accurate fatigue identification.
- **Integrated** a **WebRTC**-based browser interface with a **Python Flask** backend, enabling continuous low-latency video frame processing at **30+ FPS** and real-time alert generation.
- **Designed** an interactive monitoring dashboard to visualize drowsiness metrics and trigger automated safety alerts during simulated fatigue detection scenarios.

Multi-Disease Prediction System | *Python, Machine Learning, Scikit-learn, XGBoost*  **Oct 2025 – Nov 2025**

- **Built** an end-to-end predictive ML system for early diagnosis of multiple chronic diseases including diabetes, heart disease, and cancer using healthcare datasets.
- **Designed** a robust machine learning pipeline involving data preprocessing, feature engineering, and predictive modeling using **XGBoost**, **Random Forest**, and **SVM** algorithms.
- **Optimized** model performance through systematic feature selection and **GridSearchCV** hyperparameter tuning, achieving up to **95% prediction accuracy** while reducing false negatives.

AI Fitness Coach (Full-Stack AI Application) | *Next.js, React, TailwindCSS, Gemini API*  **Jan 2026 – Mar 2026**

- **Engineered** a full-stack AI fitness and nutrition platform integrating the **Gemini LLM API** to generate personalized workout routines and diet recommendations.
- **Implemented** real-time streaming responses and multi-session chat functionality using **React** and **localStorage**, improving perceived response time by **40%**.
- **Architected** scalable **RESTful APIs** and server-side rendered (SSR) UI components utilizing **Next.js** and **TailwindCSS** for seamless user interaction and efficient data flow.

Awards & Certifications

- **Programming in Java – NPTEL (2026):** Elite + Topper **1%**, scored **100%**; Object-Oriented Programming, Java Programming, Data Structures, Problem Solving.
- **Programming in Python – NPTEL (2024):** Top **1%**, scored **87%**; Python, Data Structures.
- **Artificial Intelligence Bootcamp – C-DAC Pune (FutureSkills PRIME):** Machine Learning, Deep Learning, Computer Vision, NLP; scored **86/100 (Very Good)**.
- **Git Training – IIT Bombay Spoken Tutorial:** Version Control using Git and GitHub; scored **80%**.

Education

B.Tech. Information Technology

Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology, Baramati

Aug 2023 – May 2027

CGPA: 8.55 / 10